HW1_314707047_連鈞宥

2026-03-04

```
library(readxl)
library(PoEdata)
```

```
#data("star5")
#head(star5)
#data("star")
#head(star)
# 找不到資料集star5 & star5_small。only star。
```

```
# 第22題
star5 <- read_excel("C:/Users/tiffa/OneDrive/Desktop/eco_fin/star5.xlsx")
head(star5)
```

```
## # A tibble: 6 × 19
##   absent  aide black   boy freelunch    id mathscore readscore regular  schid
##    <dbl> <dbl> <dbl> <dbl>     <dbl> <dbl>     <dbl>     <dbl>   <dbl>  <dbl>
## 1      5     1     1     1         0 10133       478       427       0 169280
## 2     28     1     0     0         0 10246       494       450       0 218562
## 3      2     0     1     0         1 10263       513       483       0 205492
## 4     10     0     0     1         0 10266       513       456       1 257899
## 5      3     1     0     1         0 10275       468       411       0 161176
## 6      2     0     0     1         0 10281       473       443       0 189382
## # ℹ 9 more variables: schrural <dbl>, schurban <dbl>, small <dbl>,
## #   tchexper <dbl>, tchid <dbl>, tchmasters <dbl>, tchwhite <dbl>,
## #   totalscore <dbl>, white_asian <dbl>
```

```
# a.
#total(有ABC三種可能且互斥，either A or B = not C。)
model <- lm(totalscore ~ small, data = star5, subset = (aide == 0))
summary(model)
```

```
##
## Call:
## lm(formula = totalscore ~ small, data = star5, subset = (aide ==
##     0))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -283.04  -52.94   -6.94   44.96  321.06
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  918.043      1.667 550.664  < 2e-16 ***
## small         13.899      2.447   5.681 1.44e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 74.65 on 3741 degrees of freedom
## Multiple R-squared:  0.008553,   Adjusted R-squared:  0.008288
## F-statistic: 32.27 on 1 and 3741 DF,  p-value: 1.441e-08
```

```
# b.
# read
model_read <- lm(readscore ~ small, data = star5, subset = (aide == 0))
summary(model_read)
```

```
##
## Call:
## lm(formula = readscore ~ small, data = star5, subset = (aide ==
##     0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -119.733  -21.733   -3.733   16.267  192.267
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 434.7327     0.7075 614.485  < 2e-16 ***
## small         5.8191     1.0382   5.605 2.24e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 31.68 on 3741 degrees of freedom
## Multiple R-squared:  0.008327,   Adjusted R-squared:  0.008062
## F-statistic: 31.41 on 1 and 3741 DF,  p-value: 2.235e-08
```

```
# math
model_math <- lm(mathscore ~ small, data = star5, subset = (aide == 0))
summary(model_math)
```

```
##
## Call:
## lm(formula = mathscore ~ small, data = star5, subset = (aide ==
##     0))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -163.31  -34.31   -5.31   29.69  142.69
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  483.310      1.081 447.068  < 2e-16 ***
## small          8.080      1.586   5.093  3.7e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 48.41 on 3741 degrees of freedom
## Multiple R-squared:  0.006886,   Adjusted R-squared:  0.00662
## F-statistic: 25.94 on 1 and 3741 DF,  p-value: 3.7e-07
```

```
# c.
model_aide <- lm(totalscore ~ aide, data = star5, subset = (small == 0))
summary(model_aide)
```

```
##
## Call:
## lm(formula = totalscore ~ aide, data = star5, subset = (small ==
##     0))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -283.04  -50.04   -6.70   40.64  334.64
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 918.0429     1.6129 569.174   <2e-16 ***
## aide          0.3139     2.2704   0.138     0.89
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 72.22 on 4046 degrees of freedom
## Multiple R-squared:  4.725e-06,  Adjusted R-squared:  -0.0002424
## F-statistic: 0.01912 on 1 and 4046 DF,  p-value: 0.89
```

```
# d.
model_read_aide <- lm(readscore ~ aide, data = star5, subset = (small == 0))
summary(model_read_aide)
```

```
##
## Call:
## lm(formula = readscore ~ aide, data = star5, subset = (small ==
##     0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -119.733  -21.438   -4.438   15.562  192.267
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 434.7327     0.6974 623.322   <2e-16 ***
## aide          0.7054     0.9817   0.719    0.472
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 31.23 on 4046 degrees of freedom
## Multiple R-squared:  0.0001276,  Adjusted R-squared:  -0.0001195
## F-statistic: 0.5163 on 1 and 4046 DF,  p-value: 0.4725
```

```
model_math_aide <- lm(mathscore ~ aide, data = star5, subset = (small == 0))
summary(model_math_aide)
```

```
##
## Call:
## lm(formula = mathscore ~ aide, data = star5, subset = (small ==
##     0))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -163.310  -33.919   -4.919   29.690  143.081
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 483.3102     1.0434 463.199   <2e-16 ***
## aide         -0.3915     1.4687  -0.267     0.79
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 46.72 on 4046 degrees of freedom
## Multiple R-squared:  1.756e-05,  Adjusted R-squared:  -0.0002296
## F-statistic: 0.07104 on 1 and 4046 DF,  p-value: 0.7898
```

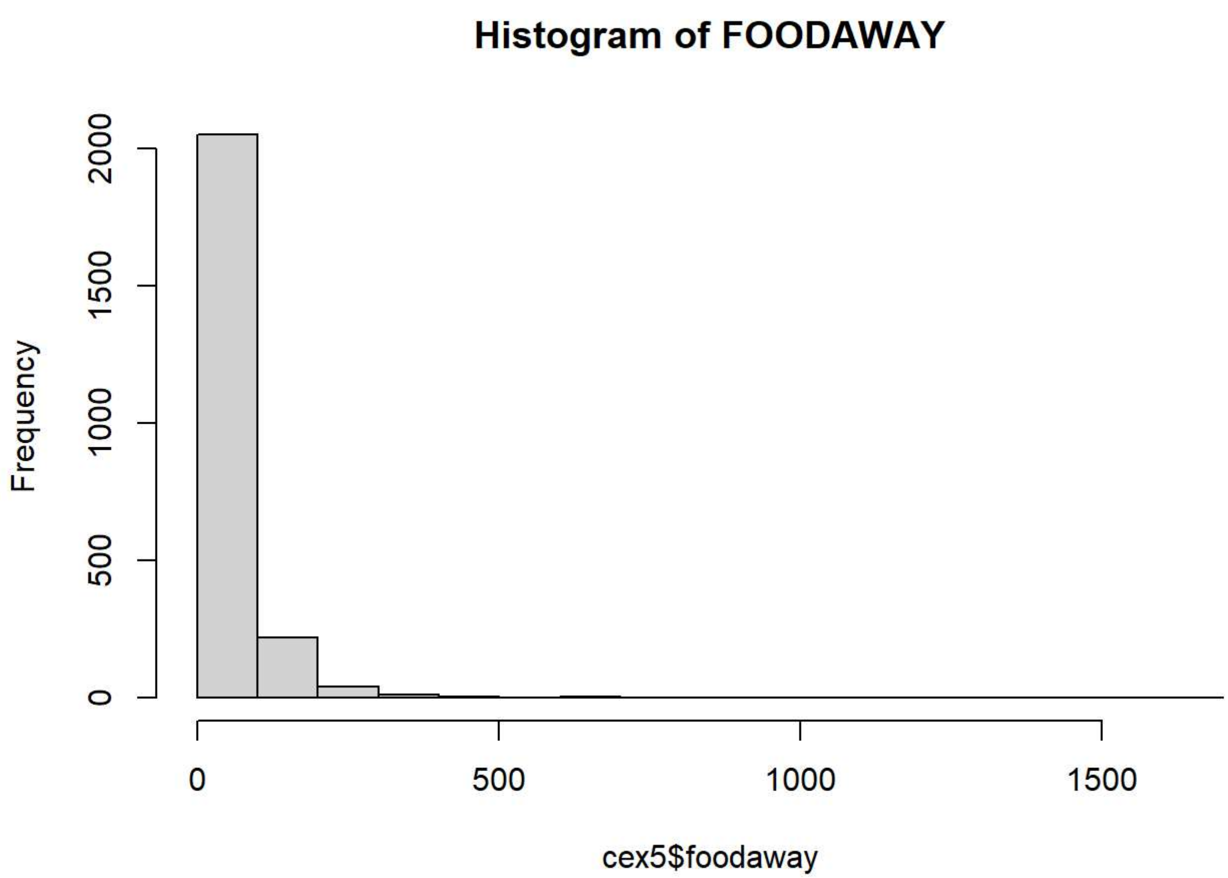
```
# 第25題
cex5 <- read_excel("C:/Users/tiffa/OneDrive/Desktop/eco_fin/cex5.xlsx")
head(cex5)
```

```
## # A tibble: 6 × 24
##       id  year month count occasion advanced alcbev appar college entert  food
##    <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl>  <dbl> <dbl>   <dbl>  <dbl> <dbl>
## 1 251162  2013     1     1        1        0   6.67  27.8       1   38   108.
## 2 251210  2013     1     1        1        0   1.67 108.        1   20.3 195.
## 3 251214  2013     1     1        1        0   0      0         0   27.2  37.8
## 4 251247  2013     1     1        1        0   0      0         1   31.7 188.
## 5 251271  2013     1     1        1        0  36.7  221.        0  305.   72.2
## 6 251326  2013     1     1        1        0  20     31.4       0  196.  260
## # ℹ 13 more variables: foodaway <dbl>, health <dbl>, hsplus <dbl>,
## #   income <dbl>, midwest <dbl>, nohs <dbl>, northeast <dbl>, older <dbl>,
## #   read <dbl>, rural <dbl>, smsa <dbl>, south <dbl>, west <dbl>
```

```
# a.
summary(cex5$foodaway)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00   14.12   33.15   51.45   69.44 1645.56
```

```
hist(cex5$foodaway, main="Histogram of FOODAWAY")
```



```r
# b.
# advanced
summary(cex5$foodaway[cex5$advanced == 1])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00   21.67   48.34   75.37   89.86 1179.00
```

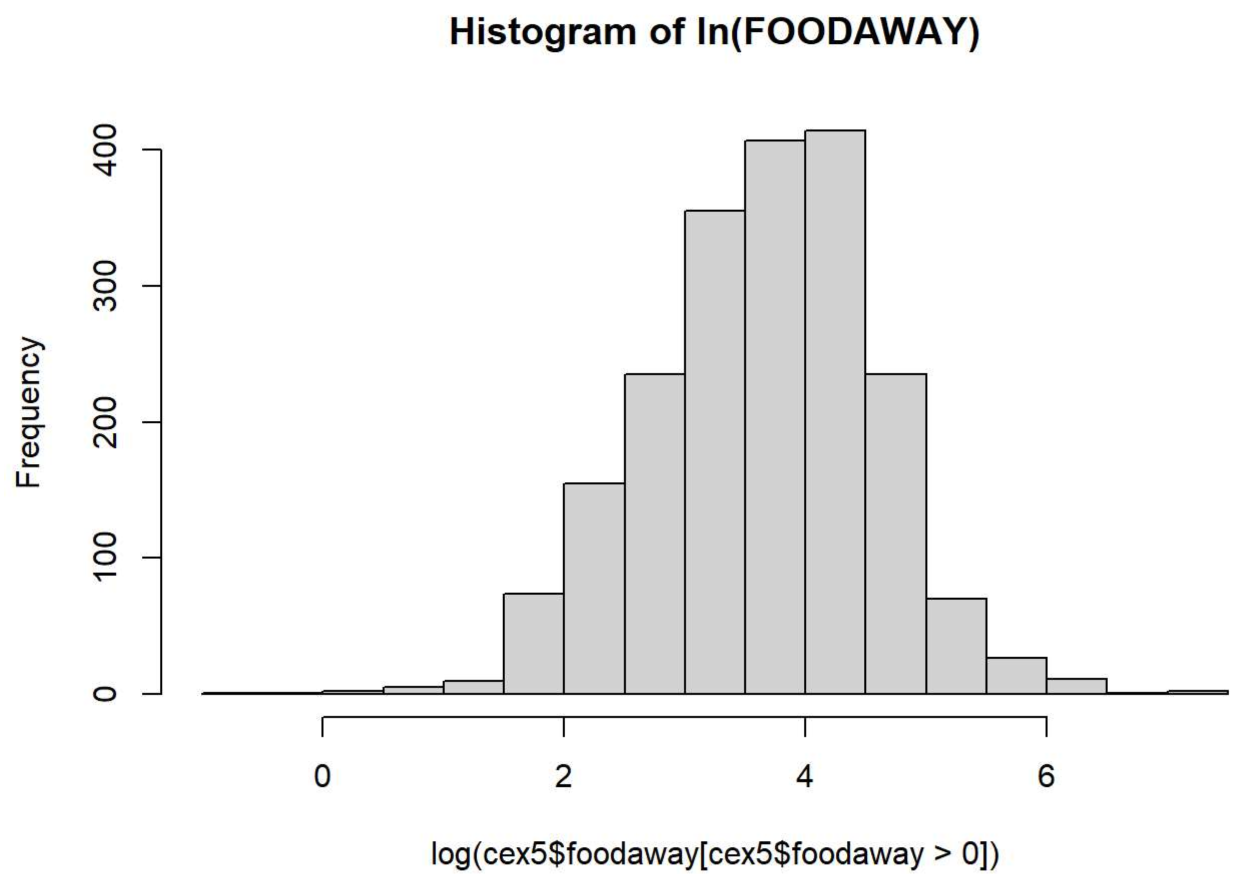
```r
# college
summary(cex5$foodaway[cex5$college == 1])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00   14.90   36.11   54.90   72.22 1645.56
```

```r
# no advanced or college
summary(cex5$foodaway[cex5$advanced == 0 & cex5$college == 0])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00    9.63   25.02   38.32   55.97  437.78
```

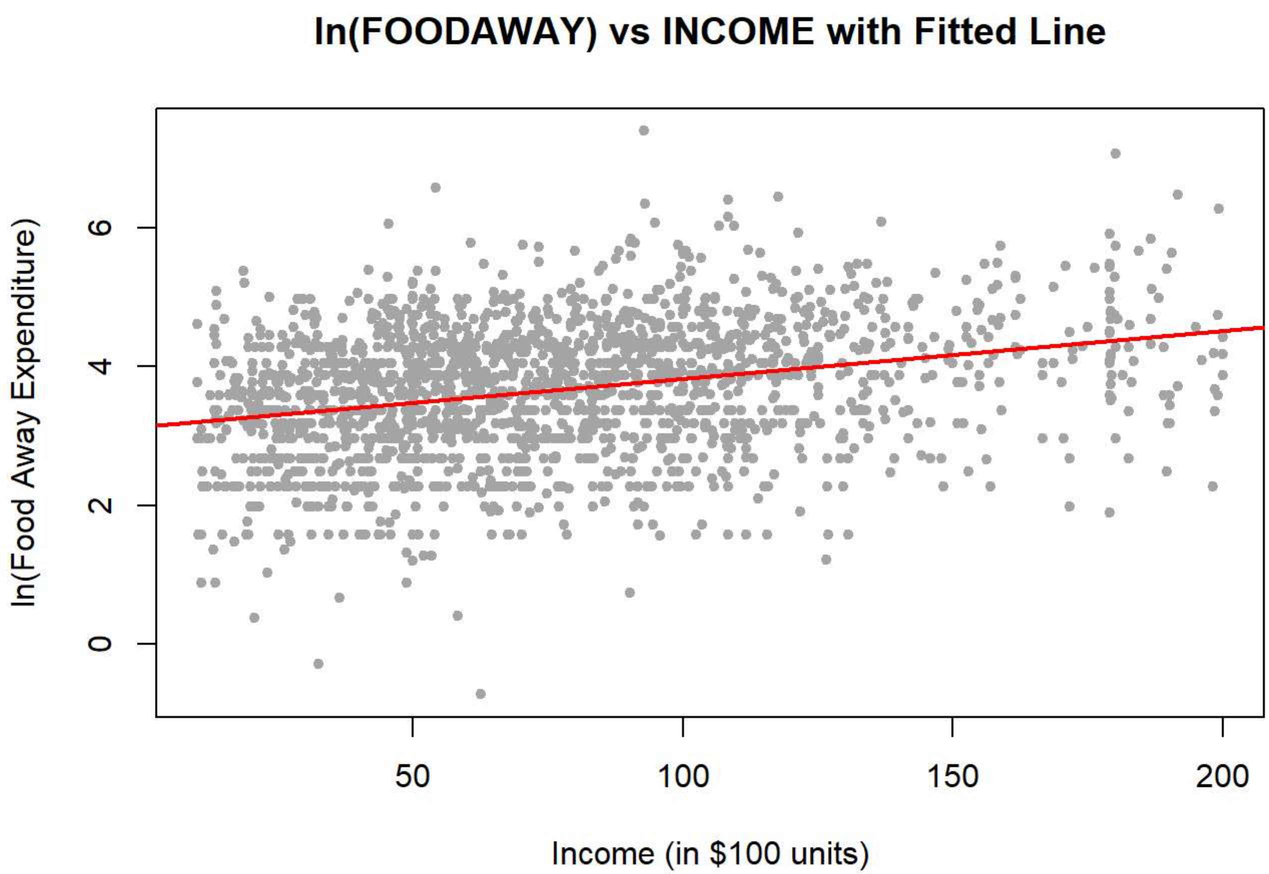
```r
# c.
hist(log(cex5$foodaway[cex5$foodaway > 0]), main="Histogram of ln(FOODAWAY)")
```



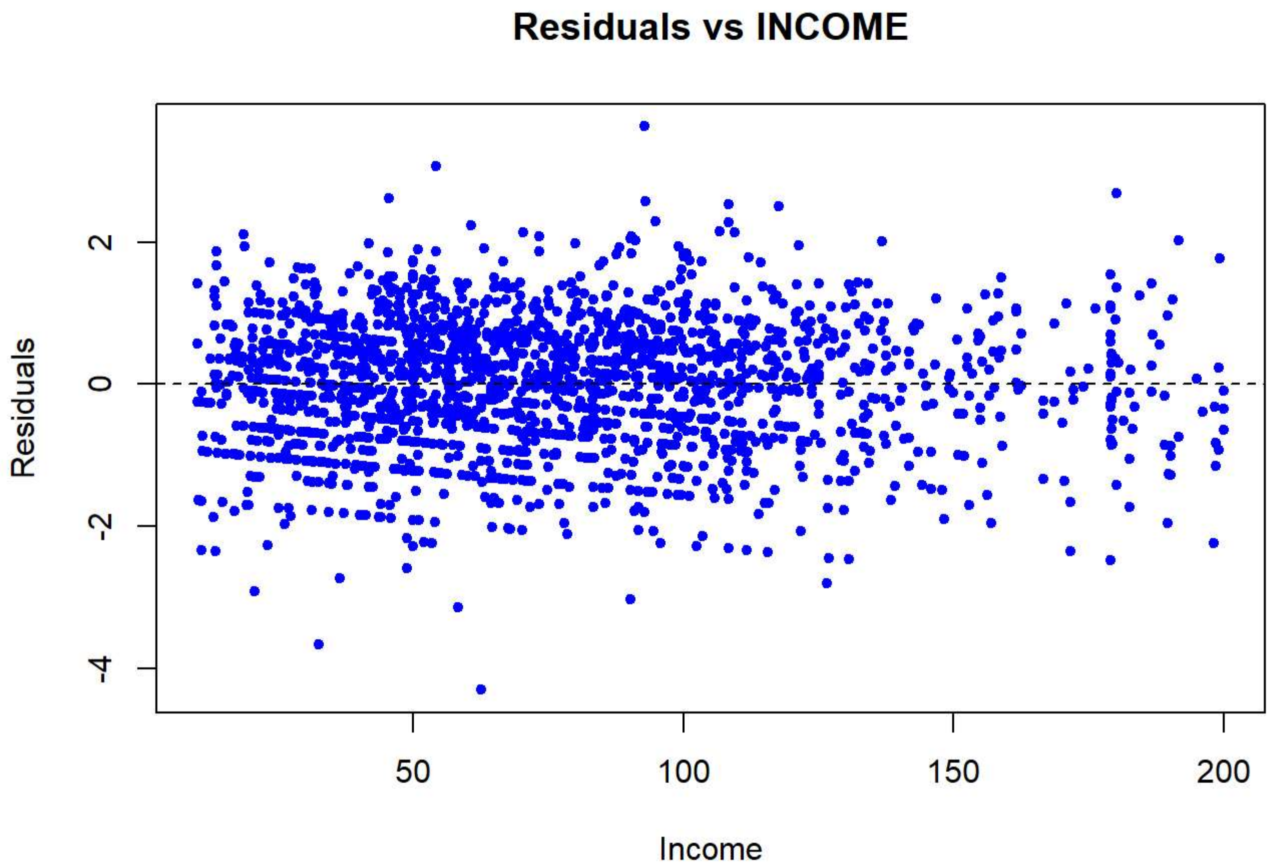

```
# d.
model_f <- lm(log(foodaway) ~ income, data = cex5[cex5$foodaway > 0, ])
summary(model_f)
```

```
##
## Call:
## lm(formula = log(foodaway) ~ income, data = cex5[cex5$foodaway >
##     0, ])
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.3034 -0.6003  0.0654  0.6119  3.6262
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.1364599  0.0420567   74.58   <2e-16 ***
## income      0.0069283  0.0004898   14.15   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.898 on 2003 degrees of freedom
## Multiple R-squared:  0.09083,    Adjusted R-squared:  0.09038
## F-statistic: 200.1 on 1 and 2003 DF,  p-value: < 2.2e-16
```

```
# e.
cex5_sub <- cex5[cex5$foodaway > 0, ]
model_d <- lm(log(foodaway) ~ income, data = cex5_sub)
plot(cex5_sub$income, log(cex5_sub$foodaway),
     main = "ln(FOODAWAY) vs INCOME with Fitted Line",
     xlab = "Income (in $100 units)",
     ylab = "ln(Food Away Expenditure)",
     pch = 20, col = "darkgrey")
abline(model_d, col = "red", lwd = 2)
```




```
# f.
e_hat <- residuals(model_d)
plot(cex5_sub$income, e_hat,
     main = "Residuals vs INCOME",
     xlab = "Income",
     ylab = "Residuals",
     pch = 20, col = "blue")
abline(h = 0, col = "black", lty = 2)
```


Residuals vs INCOME
Residuals
2
0
-2
-4
50
100
150
200
Income